

Synoptic CB: Lateral Transport Surveys SO₄/Cl

GCW Survey Samples

2025-10-08

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2025-10-03" #Date that instrument was run
Run_by = "Zoe Read" #Instrument user
Script_run_by = "Stephanie Wilson" #Code user
run_notes = " " #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = c("Raw_GCW_Data/COMPASS_2024_GCW_Lateral_Transport_Survey_Cl.txt")
raw_file_name_so4 = "Raw_GCW_Data/COMPASS_2024_GCW_Lateral_Transport_Survey_SO4.txt"

#file path and name of processed data file
processed_file_name = "Processed_GCW_Data/GCW_SO4_Cl_processed.csv"

#metadata csv
Raw_Metadata = "Raw_GCW_Data/COMPASS_SynopticCB_LatT_GCW_Discrete_Metadata.csv"

#qaqc log file path for this year
Log_path = "Raw_GCW_Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2024.csv"
```

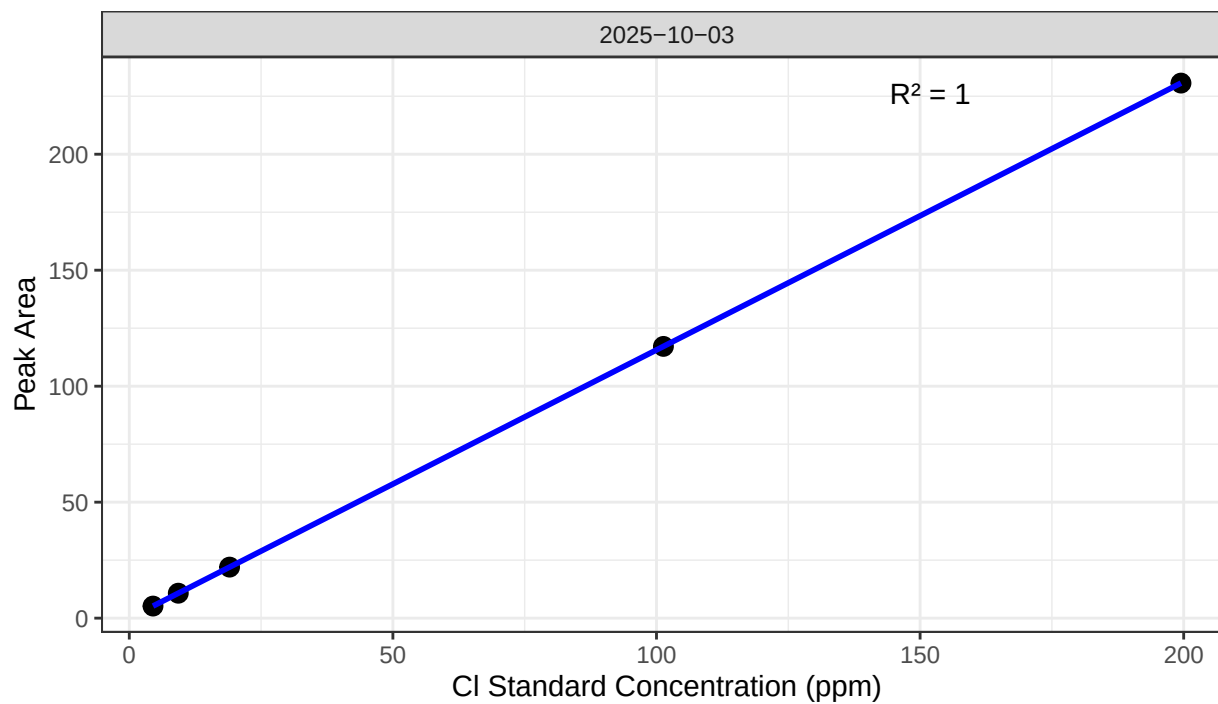
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

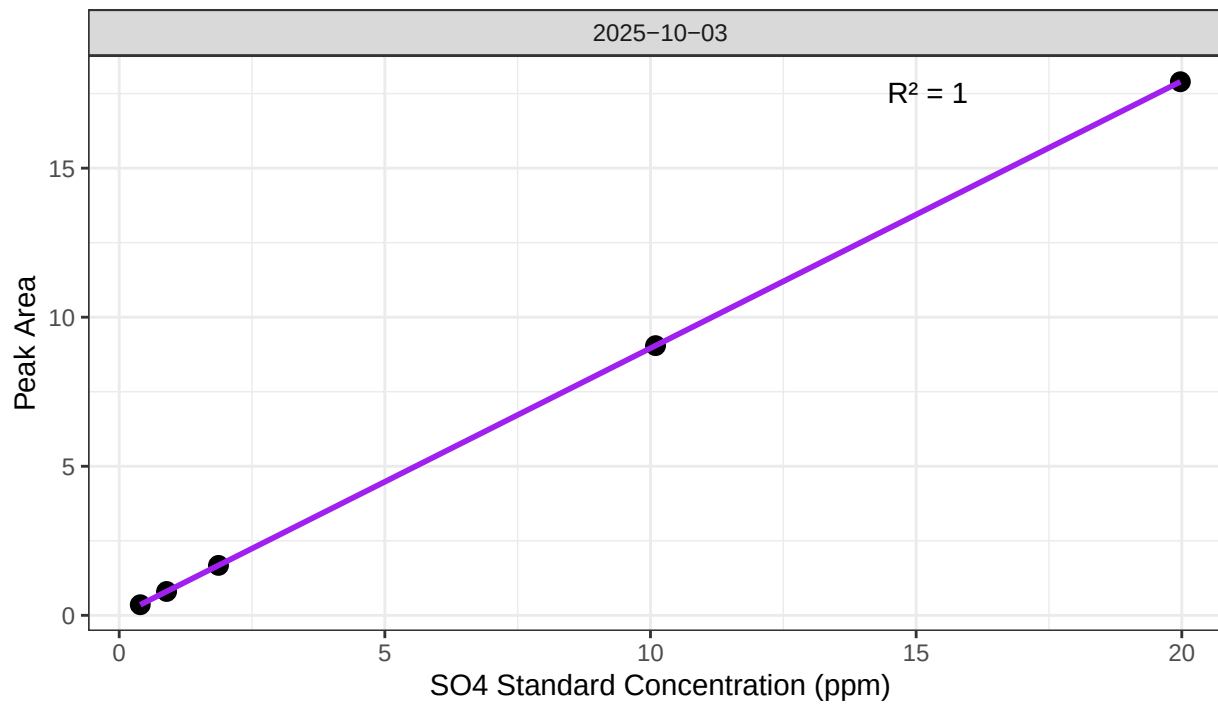
##Import Sample Data - need to change LTS to LatT and remove the date

0.2 Assess Standard Curves

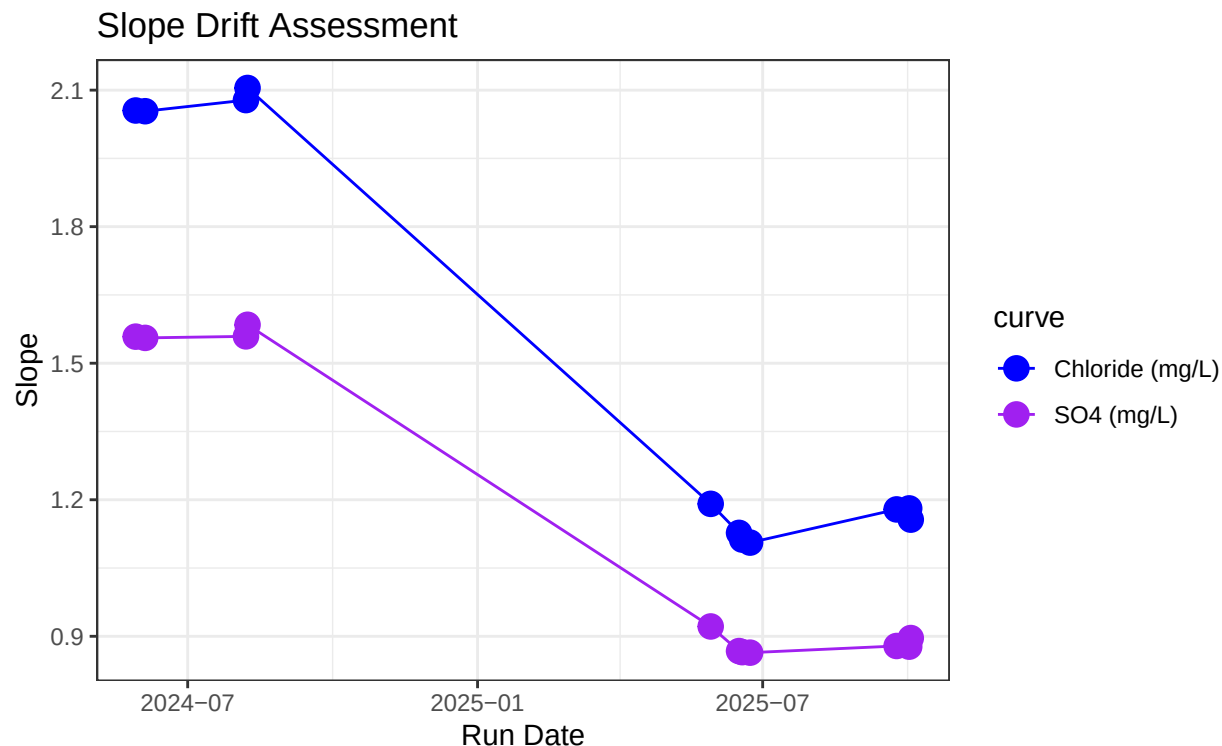
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

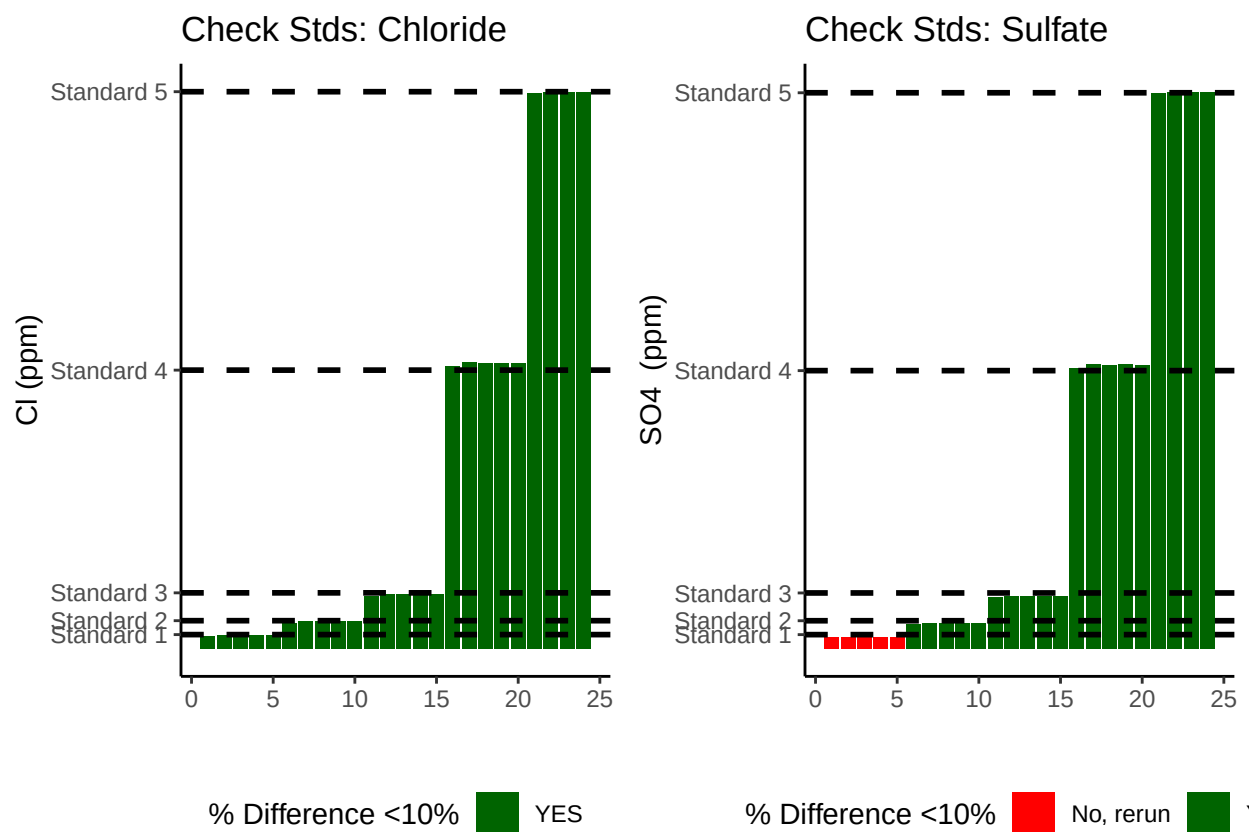
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl  cv_Cl flag_Cl
##   <chr>      <dbl> <dbl>  <dbl> <chr>
## 1 Standard 1    4.76 0.149 0.0313 Chloride Check Standard RSD within Range - P~
## 2 Standard 2    9.70 0.227 0.0234 Chloride Check Standard RSD within Range - P~
## 3 Standard 3   19.5 0.277 0.0142 Chloride Check Standard RSD within Range - P~
## 4 Standard 4  102. 0.631 0.00616 Chloride Check Standard RSD within Range - P~
## 5 Standard 5  200. 0.260 0.00130 Chloride Check Standard RSD within Range - P~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4  cv_SO4 flag_SO4
##   <chr>      <dbl>  <dbl>  <dbl> <chr>
## 1 Standard 1    0.406 0.00562 0.0139 Sulfate Check Standard RSD within Range --
## 2 Standard 2    0.906 0.00778 0.00859 Sulfate Check Standard RSD within Range --
## 3 Standard 3    1.89 0.0114 0.00604 Sulfate Check Standard RSD within Range --
## 4 Standard 4   10.2 0.0603 0.00592 Sulfate Check Standard RSD within Range --
## 5 Standard 5   20.0 0.0329 0.00164 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



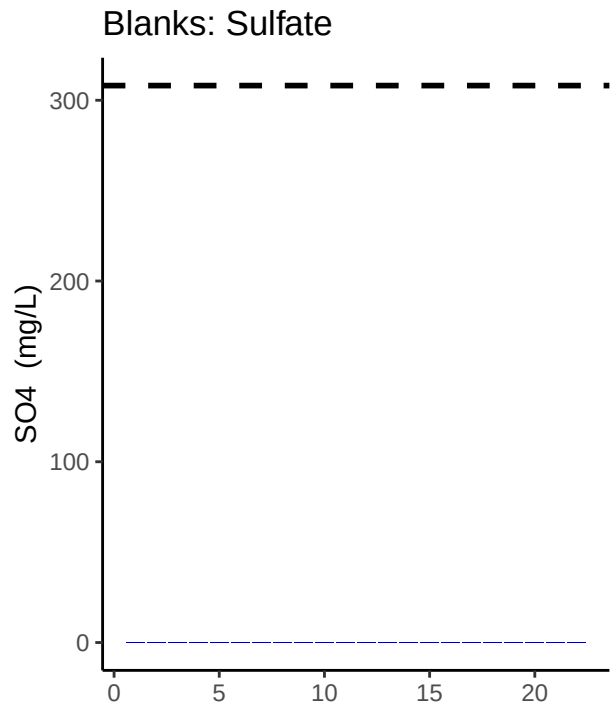
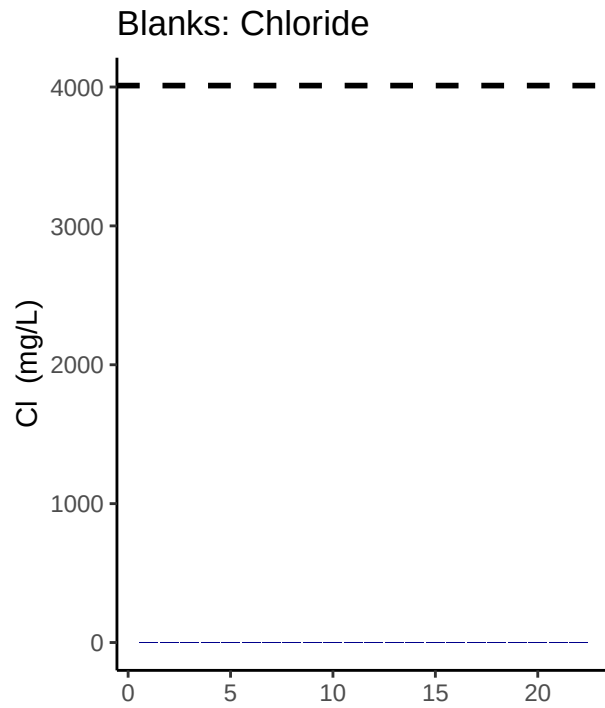
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] "<80% of Sulfate Check Standards are within range of expected concentration - REASSESS"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ Y

Blank Conc <25% Quartile Samples ■ YI

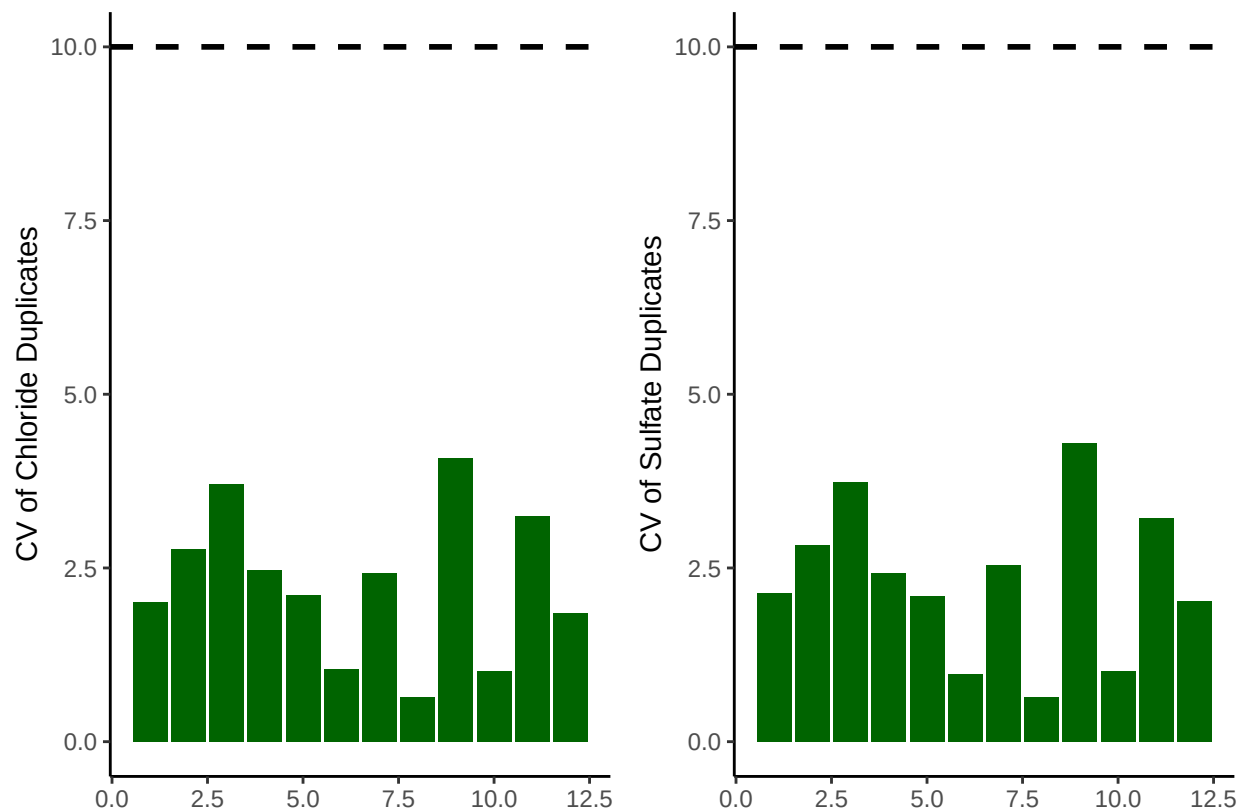
```
## Chloride blanks mean ppm:
```

```
## [1] 0.0004181818
```

```
## Sulfate blanks mean ppm:
```

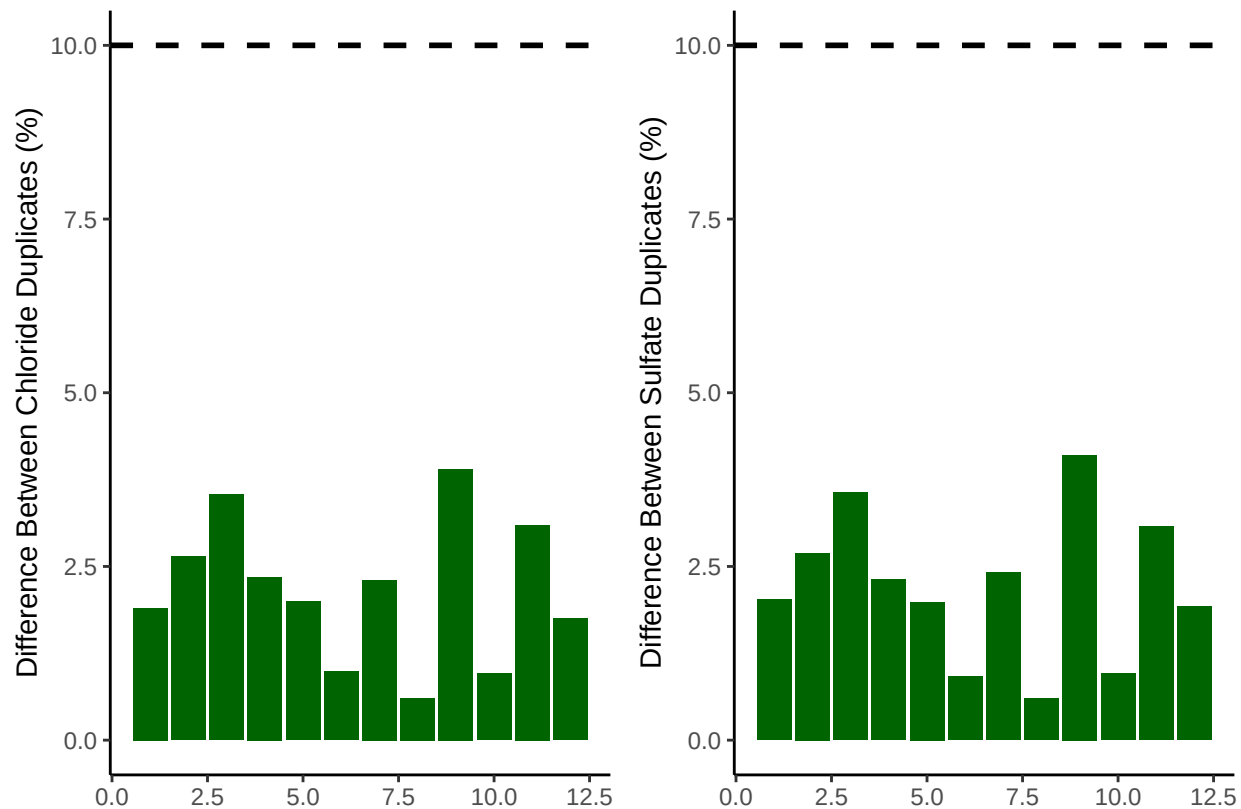
```
## [1] 0
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# = ((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000
```

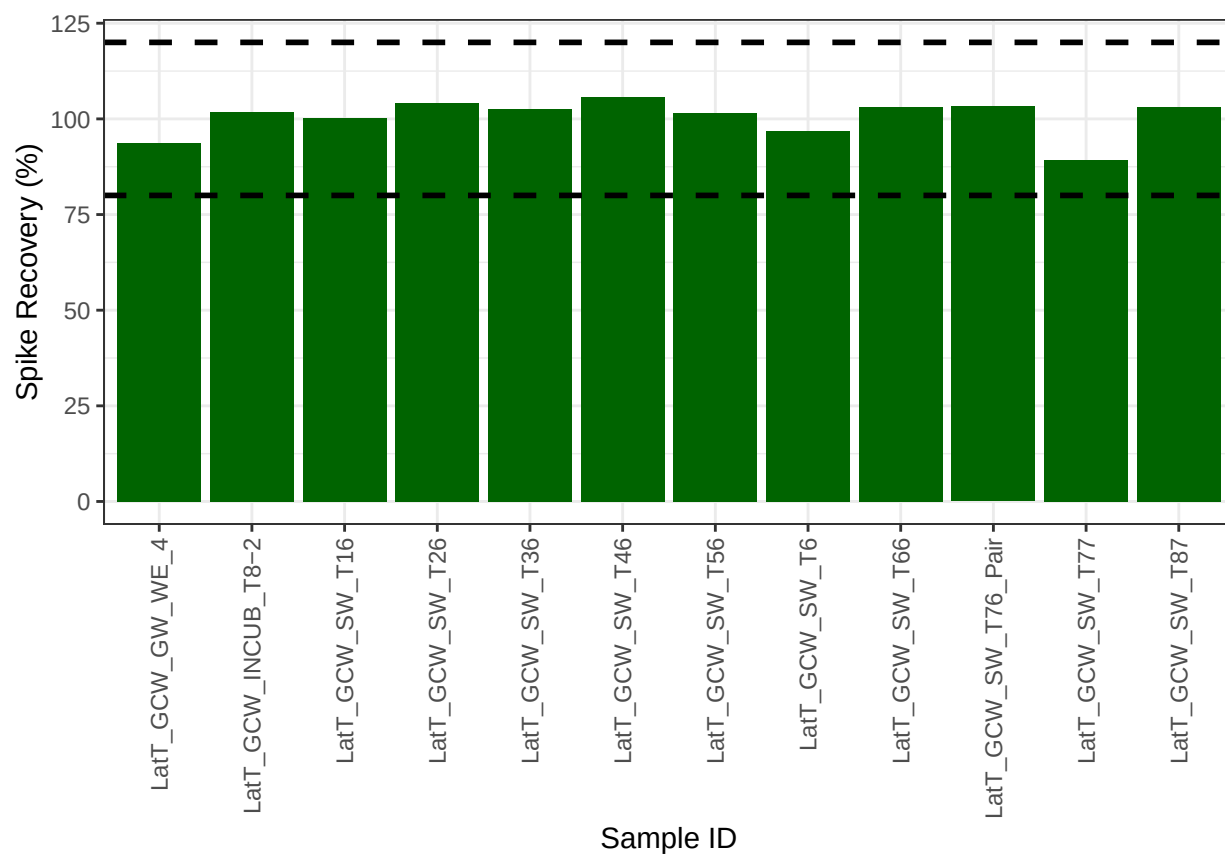
0.7 Check samples with metadata

```
## Some sample IDs are missing from metadata.
```

```
## [1] "LatT_GCW_GW_TR_1"
```

0.8 Add in dilution factors

0.9 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.10 Check if samples within the range of the standard curve

```
## Sample Flagging
```

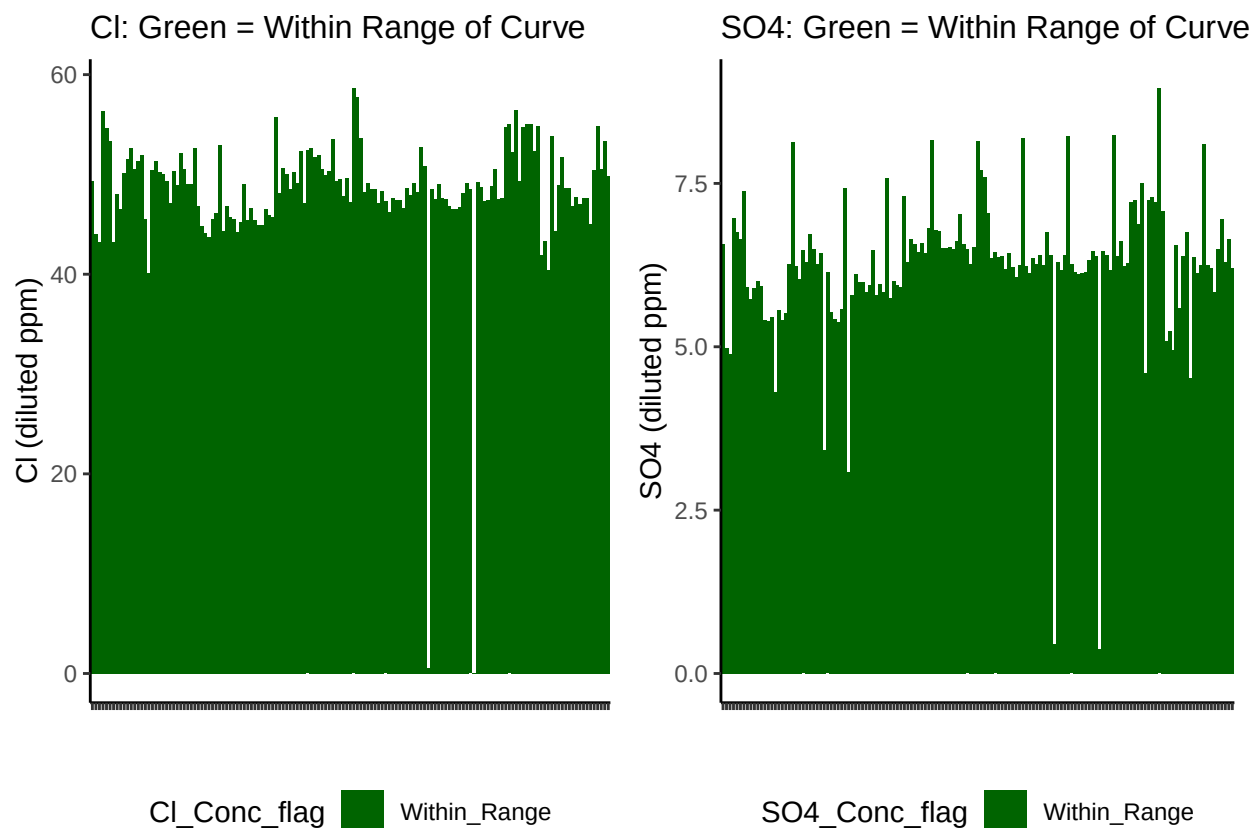


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

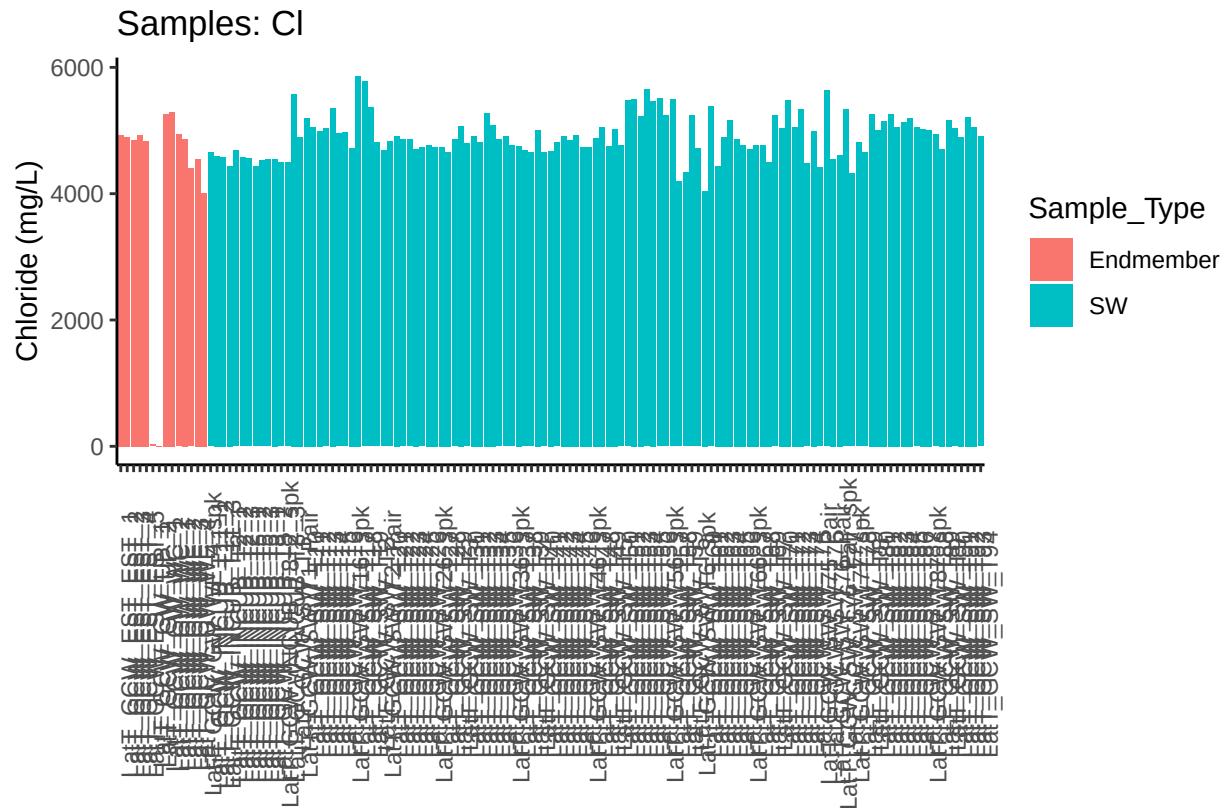
Table 2: Cl samples

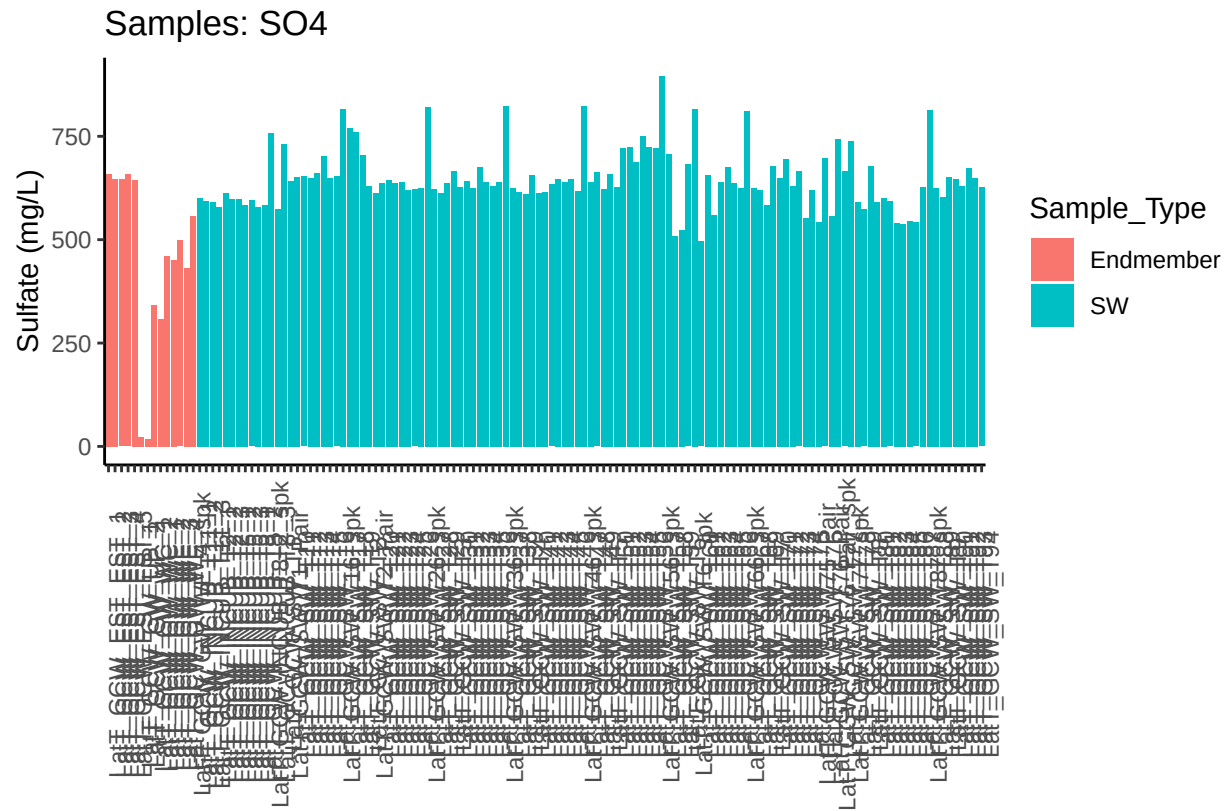
Cl_Conc_flag	Percent_samples
Within_Range	100

0.11 Visualize Data

```
## Visualize Data
```

```
## Warning in rbind(c("LatT", "GCW", "EST", "EST", "1"), c("LatT", "GCW", "EST", :  
## number of columns of result is not a multiple of vector length (arg 1)
```





0.12 Export Processed Data

#end